Practice Quiz: Systems

Part A: Multiple Choice

1. A crocheter makes a flat circle using 6 increases per round in single crochet. This fabric has: A) Positive Gaussian curvature — it cups into a bowl B) Negative Gaussian curvature — it ruffles and waves C) Zero Gaussian curvature — it lies flat on the table D) Variable curvature — it alternates between flat and curved

2. Daina Taimina's hyperbolic crochet models are created by: A) Using fewer increases than needed per round, causing the fabric to cup B) Using exactly the right number of increases, keeping the fabric flat C) Using more increases than needed per round, causing the fabric to ruffle D) Crocheting in rows instead of rounds to create straight edges

3. In a crocheted fabric viewed as a graph, each stitch is a **node** and each connection between stitches is an **edge**. When you work a shell stitch (5 dc in 1 st), the base stitch becomes: A) A node with very low degree, since the shell replaces it B) A node with high degree, since five new stitches connect to it C) An edge, not a node D) A disconnected component of the graph

4. The key structural difference between crochet worked in rows and crochet worked in the round is: A) Rows use more yarn than rounds B) Rows create a feedforward (acyclic) graph; rounds create a recurrent (cyclic) graph C) Rows have higher Gaussian curvature than rounds D) Rounds cannot create flat surfaces

5. A spike stitch, where the hook reaches past the previous row into a stitch two or more rows below, is analogous in neural network terms to: A) A bias unit B) A skip connection (residual connection) that bypasses intermediate layers C) An output neuron D) A dropout layer

6. The Euler characteristic of a closed crocheted sphere (like an amigurumi ball) is: A) 0 B) 1 C) 2 D) -1

7. In Active Inference, the Markov blanket of a crochet project is the working edge — the live loops and hook. When crocheting in the round, this Markov blanket is: A) An open curve with two endpoints B) A closed curve forming a continuous loop C) A point with no extent D) Identical to the blanket in flat row-by-row crochet

Part B: Short Answer

1. Explain the relationship between the number of increases per round in a crocheted circle and the Gaussian curvature of the resulting fabric. Give three specific examples: one that produces zero curvature, one that produces negative curvature, and one that produces positive curvature. For each, describe what the fabric looks like physically.

2. Crocheted fabric can be viewed as a layered network graph, similar to a neural network. Describe how the foundation chain, subsequent rows, and stitch connections map onto the concepts of input layer, hidden layers, and weighted connections. What would a "skip connection" look like in crochet, and how is it physically made?

3. A crocheter is working a flat circle and notices that the edges are beginning to ruffle after round 8. Using the framework of Active Inference, describe what has happened in terms of the generative model, prediction error, and the crocheter's options for minimizing free energy. Connect your answer to the topological concept of curvature.